The Hexagonality Law in Real Materials II

Entropy and pressure

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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. The Hexagonality Law (#66) says that in ideal stackings every kind of piece is a straight line in the share h of hexagonal layers; in the physicists' layer model this is a property with only a nearest term (#68). DICTO proposed to test the law's form on two quantities that decide which stacking a crystal takes: its entropy and its enthalpy under pressure. **Entropy:** Mau and Huse wrote the stacking entropy of hard spheres in the same layer model; at close packing their nearest term is 55 and the next 6 (in 10^{-5} k_B per sphere), so the entropy is about 90% of the law's form, and the farther terms grow toward melting (37, 18 and 9). **Pressure:** we prove that if a crystal's volume follows the law's form, pressure shifts only the balance between cubic and hexagonal layers and never reorders the longer stackings among themselves, and that it keeps the fault ratios 2 : 2 : 1. From published volumes and energies, at first order, pressure strengthens cubic 3C silicon carbide against 6H, 4H and 2H, as experiments find (6H turns into 3C above about 4.5 GPa); in silicon and germanium, whose volumes bend away from the straight line, pressure tilts them toward hexagonal-type stackings, mainly through the farther term.

1. Two quantities that choose the stacking

Which stacking a crystal takes is decided by its free energy: energy, entropy, and, under pressure, the term pressure × volume. Paper #68 compared real volumes with the law's straight line. DICTO proposed to go further and test the law's form on the two other quantities: the entropy of hard spheres, where energy plays no part at all, and the enthalpy under pressure, where the volume enters directly.

2. The layer model

As in #68: in the axial Ising (ANNNI) model a property X of a stacking, per atom pair and relative to 3C, is $X(2H) = 2(J_1 + J_3)$, $X(4H) = J_1 + 2J_2 + J_3$, $X(6H) = (2/3)(J_1 + 2J_2 + 3J_3)$ (convention of Keller et al., 2023), and X is a straight line in h exactly when $J_2 = J_3 = 0$ (#68, Proposition). Mau and Huse (1999) use the same idea with the layer types as spins: a field h for each layer, a coupling J between neighbouring layers, and h′, J′ for longer ranges. A straight line in the hexagonality is the case of the field alone.

3. Entropy of hard spheres

Hard spheres have no energy; their stackings differ only in entropy, by about 10^{-3} k_B per sphere, with cubic stacking the highest (fcc − hcp = 0.00115 k_B per sphere at close packing,

Mau and Huse; 0.001164(8), Elser 2014). Mau and Huse measured the layer couplings of this entropy:

Density	field h	coupling J	h′	J/h
close packing	55	6	—	0.11
near melting (about 74% of close packing)	37	18	9	0.49

(in 10^{-5} k_B per sphere; all terms favour cubic stacking). So at close packing the stacking entropy is about 90% of the law's form, a straight line in the hexagonality, and the farther terms grow as the spheres rattle more freely toward melting. Elser points out why the entropy differs at all: if every sphere moved in a fixed cage of its neighbours, the difference between cubic and hexagonal stacking would be exactly zero; it comes from collective motion. The law describes the fixed geometry; the entropy departs from its form only as far as the spheres move together. Entropy is not volume, and we compare the form only, not the numbers.

4. Pressure

At pressure P a stacking's enthalpy per pair is E + PV, so in the layer model the volume couplings J_k^V add $P{\cdot}J_k^V$ to the energy couplings.

> **Proposition.** If a crystal's volume per pair follows the law's form (a straight line in the hexagonality, $J_2^V = J_3^V = 0$), then at first order pressure changes only J_1: it shifts the balance between cubic and hexagonal layers, adds the same amount per hexagonal layer to every stacking, keeps the fault ratios intrinsic : extrinsic : twin of its own contribution at 2 : 2 : 1, and leaves J_2 and J_3, which order the longer stackings among themselves, unchanged.

Proof. The volume term is P·V, and V has couplings $(J_1^V, 0, 0)$ by assumption; so P·V adds $P{\cdot}J_1^V$ to J_1 and nothing to J_2, J_3. A property with nearest term only is s·h with $s = 2J_1$ (#68), the same amount per hexagonal layer, and its fault values are $4J_1$, $4J_1$, $2J_1$ (#68). ∎

Real crystals are not exactly of this form, so we computed, at first order (couplings and volumes at zero pressure), the margin by which cubic 3C beats 6H, 4H and 2H, from the same published data as #68 (silicon carbide with van der Waals correction: energies 15, 2.7 and 1.3 meV per pair above 3C for 2H, 4H and 6H, Kawanishi and Mizoguchi; silicon and germanium, Keller et al.).

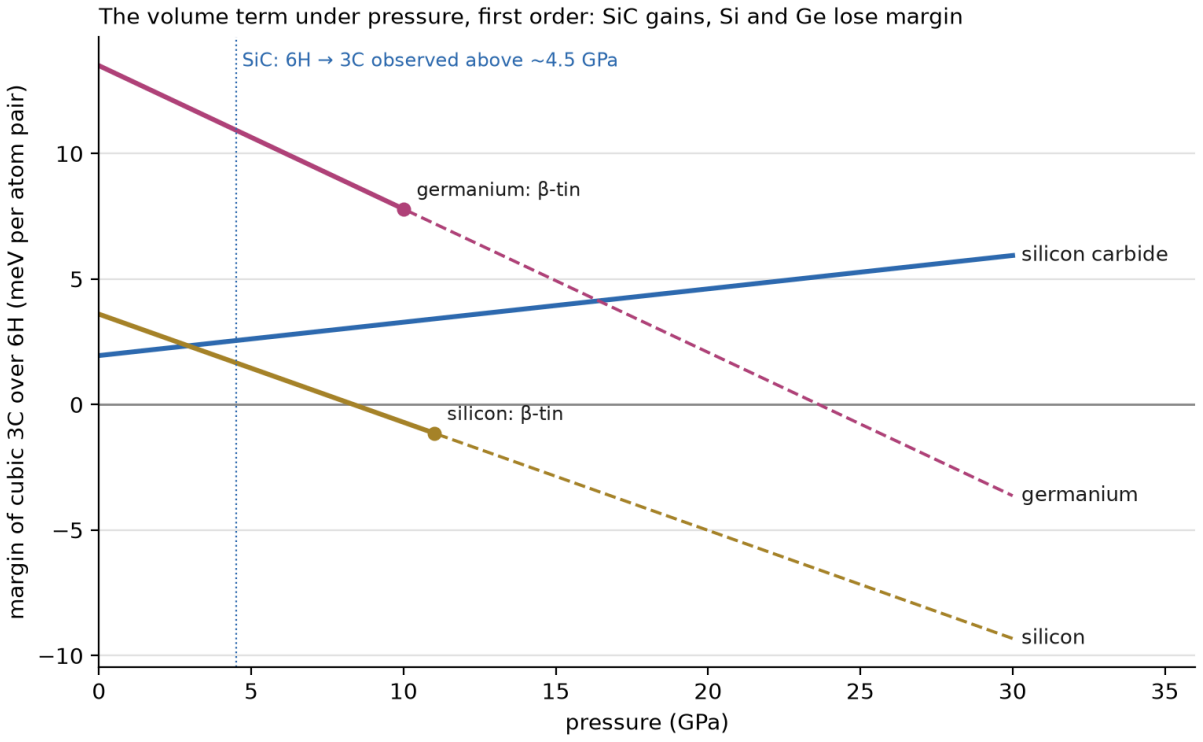


Figure 1. The margin of cubic 3C over 6H, energy plus pressure × volume per atom pair, against pressure, at first order. Silicon carbide gains margin; silicon and germanium lose it; their lines are dashed beyond their own transitions to β-tin (near 11 and 10 GPa). The dotted line marks the pressure above which 6H silicon carbide is observed to turn into 3C.

Material	nearest share of the volume couplings	3C over 6H	3C over 4H	3C over 2H
silicon carbide	0.78	+0.13	+0.11	+0.17
silicon	0.39	−0.43	−0.36	−0.13
germanium	0.70	−0.57	−0.50	−0.32

(change of the margin in meV per atom pair per GPa; nearest share = $|J_1^V| / (|J_1^V| + |J_2^V| + |J_3^V|)$.) Silicon carbide's volume is close to the law's form and rises slightly with the hexagonality, so pressure strengthens the cubic stacking. That agrees with experiment: 3C becomes more stable with pressure, and 6H turns into 3C above about 4.5 GPa at high temperature (Sugiyama and Togaya, 2001; review, 2018). In silicon and germanium the volume falls with the hexagonality and bends away from the straight line, and pressure tilts them toward hexagonal-type stackings; at first order silicon's margin over 6H would close near 8 GPa, but silicon turns into β-tin near 11 GPa and the first-order estimate ignores how compression changes the couplings, so this is a tilt, not a predicted phase. For germanium the margins close only at 24–50 GPa, far beyond its own transition.

5. Conclusion

The law's straight line describes the ideal part of what chooses a stacking. Near close packing the entropy of hard spheres is about 90% of that form. Under pressure, a volume of that form could only tilt the balance between cubic and hexagonal; silicon carbide, close to it, is pushed toward cubic as experiments see; silicon and germanium, further from it, are tilted the other way, through the farther couplings their relaxation adds.

6. Prior art and credit

The layer (ANNNI) description of polytypes is Heine's; the entropy couplings of hard spheres are Mau and Huse's (1999), with Bolhuis, Frenkel and others before them; the net entropy difference and its phonon origin are Elser's (2014); the energies and volumes are those of Kawanishi and Mizoguchi (2015) and Keller et al. (2023); the high-pressure behaviour of silicon carbide is from Sugiyama and Togaya (2001) and a 2018 review. The statement that a volume of the law's form shifts only J_1 under pressure is an immediate consequence of the model, which we state with its proof. The Hexagonality Law is #66; real polytypes are #68.

> **Sources checked** (28 October 2026). Read: S.-C. Mau and D. A. Huse, Stacking entropy of hard-sphere crystals, Phys. Rev. E 59, 4396 (1999), arXiv cond-mat/9810287; V. Elser, Phonon contribution to the entropy of hard-sphere crystals, arXiv 1403.1784 (2014); S. Kawanishi and T. Mizoguchi, arXiv 1512.04626 (energies and Table I); M. Keller et al., arXiv 2302.10822 (Tables I and II); all in the reference library. Search summaries, not read in full: Y. Sugiyama and M. Togaya (2001, Hokkaido University pdf); the 2018 review of silicon carbide at high pressure and temperature (Daviau and Lee, Yale pdf); Karch et al., Phys. Rev. B (1996); Maitani et al. (2024). Reference library (ask-library: "hard sphere crystal fcc hcp entropy difference stacking"; "Mau Huse stacking entropy hard-sphere crystals"; "Bolhuis Frenkel hard sphere fcc hcp free energy"): Ganesh and Nasr (2023), Hart, Hansen and Kuhs (2021), others not relevant.

> **Data and code.** Studies 2026-10-28-hard-sphere-entropy (README) and 2026-10-28-pressure (pressure.py, pressure.json, fig70.py), attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; Real Polytypes and the Hexagonality Law (#68), 10.5281/zenodo.23719275.